# Sampling Distributions & Central Limit Theorem

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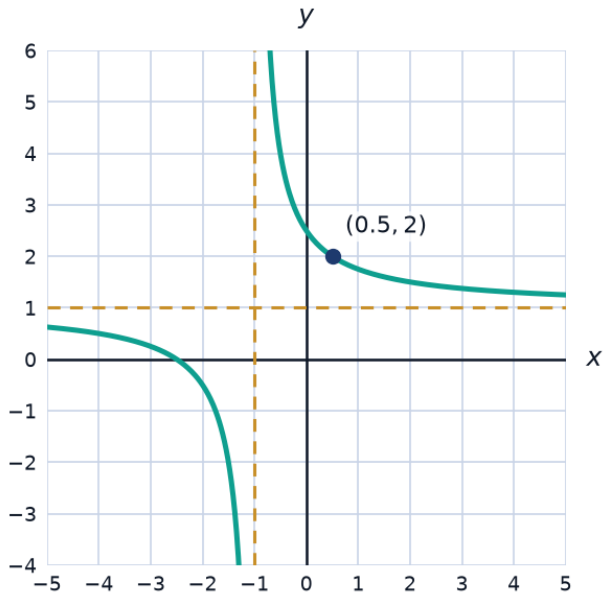
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: What is a sampling distribution?

Explain the difference between a population distribution and the sampling distribution of a sample mean $\bar{x}$.

**2** [Foundational] • Target: What is a sampling distribution?

Explain why the sampling distribution of $\bar{x}$ has less variability than the original population distribution.




**3** [Foundational] • Target: Mean and standard deviation of the sampling distribution

A population has mean $\mu = 60$ and standard deviation $\sigma = 12$. For samples of size $n = 36$, find the mean and standard deviation of the sampling distribution of $\bar{x}$.

**4** [Foundational] • Target: Mean and standard deviation of the sampling distribution

Using the same population, find the standard error for a larger sample of size $n = 144$, and explain what happens to the standard error as sample size increases.

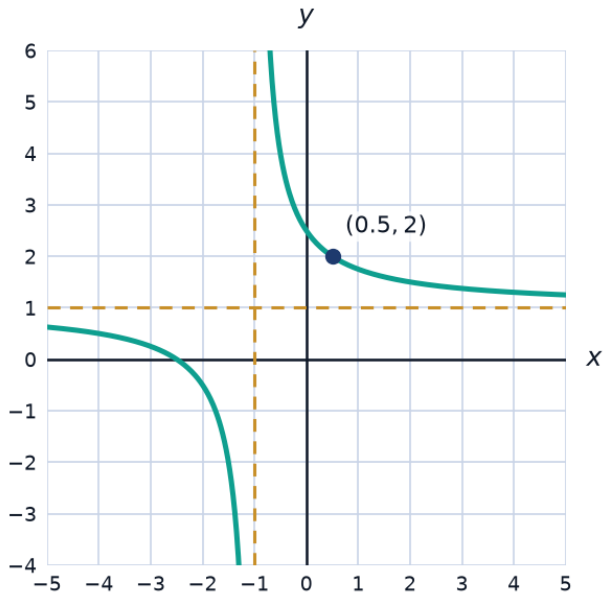
**5** [Foundational] • Target: The Central Limit Theorem

State the Central Limit Theorem in your own words, including what happens to the shape of the sampling distribution of $\bar{x}$ as sample size $n$ increases.

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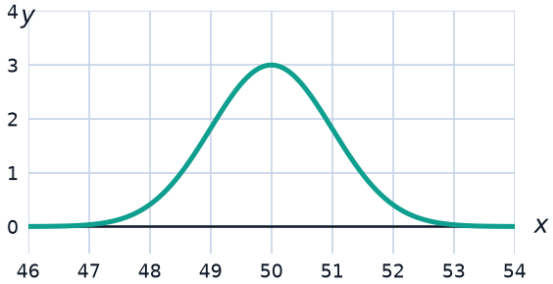
**6** [Foundational] • Target: The Central Limit Theorem

Explain why the Central Limit Theorem is considered one of the most important results in statistics, referencing its practical usefulness.




**7** [Foundational] • Target: Applying the Central Limit Theorem

A population of package weights has mean $\mu = 50$ lbs and standard deviation $\sigma = 8$ lbs (population shape unknown). For a random sample of $n = 64$ packages, find the probability the sample mean weight exceeds $52$ lbs, given $P(Z > 2) \approx 0.0228$.




## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Sampling distribution of a sample proportion

A population has a true proportion $p = 0.4$ favoring a proposal. For samples of size $n = 100$, find the mean and standard deviation of the sampling distribution of the sample proportion $\hat{p}$, using $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.

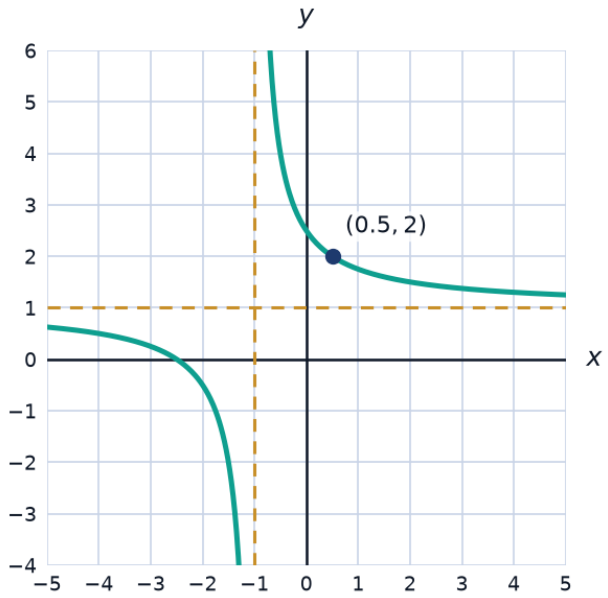
**9** [Intermediate] • Target: Sampling distribution of a sample proportion

State the two conditions typically required to treat the sampling distribution of $\hat{p}$ as approximately normal.

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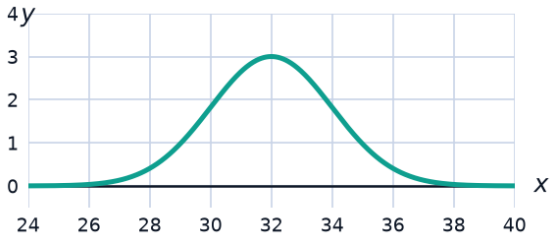
**10** **[Intermediate]** • **Target: Comparing sampling variability across sample sizes**

Explain, in the context of opinion polling, why a poll of 1000 people gives a more reliable estimate of a population proportion than a poll of 50 people, connecting your answer to the formula for $\sigma_{\hat{p}}$.




**11** **[Intermediate]** • **Target: Applying the sampling distribution of the sample mean**

A population of commute times has mean $\mu = 32$ minutes and standard deviation $\sigma = 10$ minutes. For a random sample of $n = 25$ commuters, find the probability the sample mean commute time exceeds 35 minutes, given $P(Z > 1.5) \approx 0.0668$.




**12** **[Intermediate]** • **Target: Sample size and the CLT threshold**

A population of insurance claim amounts is strongly right-skewed. Explain why a sample of size $n = 15$ might not be large enough to safely use CLT-based normal methods, but a sample of size $n = 200$ likely would be.

**13** **[Intermediate]** • **Target: Standard error vs. standard deviation**

Explain the difference between the standard deviation of an individual population value and the standard error of a sample mean $\bar{x}$.

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**14** [Intermediate] • Target: Standard error vs. standard deviation

A quality control engineer at a bottling plant knows that individual bottle fill volumes are approximately normal with mean $500$ mL and standard deviation $8$ mL. Every hour, the engineer measures a sample of 16 bottles and computes the sample mean fill volume. Describe the shape, mean, and standard deviation of the sampling distribution of this hourly sample mean, and explain whether the Central Limit Theorem's usual $n \geq 30$ guideline is even needed to justify treating this sampling distribution as normal.

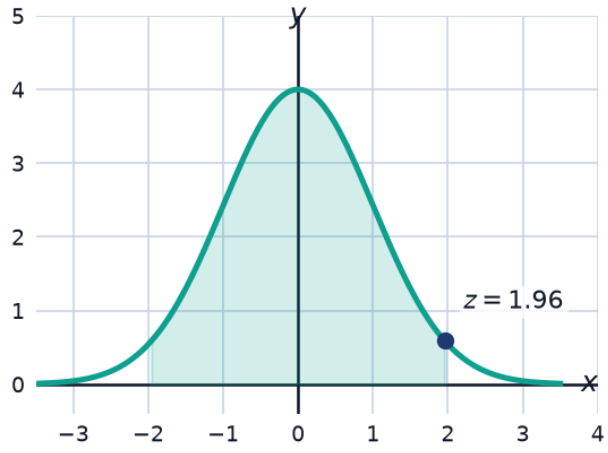
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A political pollster samples 900 voters and calculates a sample proportion $\hat{p} = 0.53$ supporting a ballot measure, from a true population proportion of $p = 0.50$. Verify that both conditions ($np \geq 10$ and $n(1-p) \geq 10$) are satisfied for treating the sampling distribution of $\hat{p}$ as approximately normal, then find the standard deviation of this sampling distribution, to four decimal places.

**16** [Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 2.01 Sampling Methods & Bias

A large factory produces light bulbs whose individual lifespans are strongly right-skewed (most bulbs fail early, but a few last an unusually long time), with population mean $\mu = 800$ hours and standard deviation $\sigma = 250$ hours. An engineer takes a random sample of $n = 64$ bulbs and computes the sample mean lifespan. Explain, step by step, why it is reasonable to treat this sampling distribution of $\bar{x}$ as approximately normal despite the population itself being skewed, then compute the standard error of $\bar{x}$.




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**17** [Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

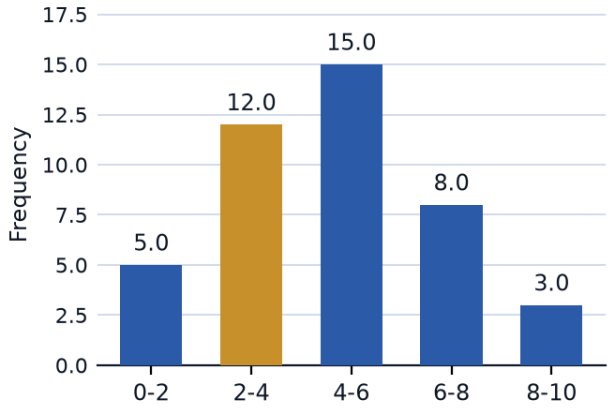
An online retailer knows that 22% of all website visitors make a purchase. On a particular day, the retailer records the browsing sessions of a random sample of 500 visitors and calculates the sample proportion who made a purchase. Verify the two conditions needed to treat this sampling distribution as approximately normal, then compute the standard deviation of the sampling distribution of the sample proportion, to four decimal places, and describe what this standard deviation represents in the context of comparing many different days' worth of visitor samples.

**18** [Challenging] • Target: Sampling distribution of a sample proportion • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

The distribution of cholesterol levels in a population is approximately normal with $\mu = 190$ mg/dL and $\sigma = 24$ mg/dL. (a) What proportion of individuals have cholesterol levels above 226 mg/dL? (b) What cholesterol level marks the 90th percentile?

**19** [Challenging] • Target: Comparing sampling variability across sample sizes • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A histogram is used to display the distribution of the number of hours college students reported spending on social media per day. The bins for the histogram are as follows: 0-2 hours, 2-4 hours, 4-6 hours, 6-8 hours, and 8-10 hours. The heights of the bars are 5, 12, 15, 8, and 3, respectively. What is the total number of students represented by this histogram?




**20** [Challenging] • Target: Applying the sampling distribution of the sample mean • Cross-Lesson: 3.03 The Normal Distribution

A coffee brand claims its jars contain a mean of 200 grams of coffee. A consumer agency takes an SRS of 16 jars, finding $\bar{x} = 197.5$ g and $s = 4.0$ g. Assume jar weights are approximately normally distributed. Conduct a significance test of $H_0 : \mu = 200$ vs. $H_a : \mu < 200$ at $\alpha = 0.05$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: What is a sampling distribution?

A population has mean $\mu = 100$ and standard deviation $\sigma = 60$. For samples of size $n = 100$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

**a.** 6

**b.** 5

**c.** 7

**d.** 60

**22** [Foundational] • Target: Mean and standard deviation of the sampling distribution

A population has mean $\mu = 100$ and standard deviation $\sigma = 117$. For samples of size $n = 169$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

**a.** 10

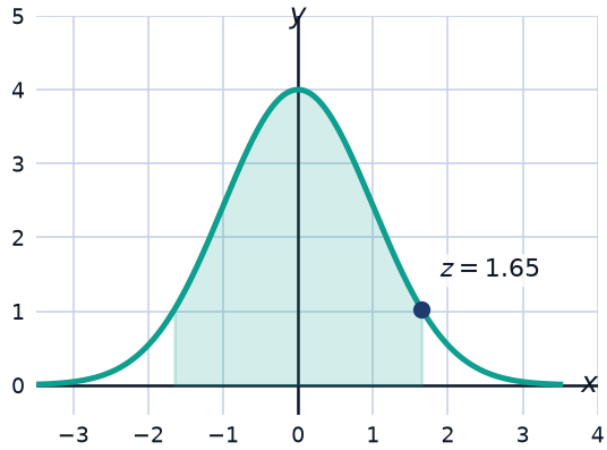
**b.** 117

**c.** 9

**d.** 8

**23** [Foundational] • Target: The Central Limit Theorem

A population has mean $\mu = 100$ and standard deviation $\sigma = 52$. For samples of size $n = 169$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.




**a.** 4

**b.** 5

**c.** 3

**d.** 52

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**24** [Foundational] • Target: Applying the Central Limit Theorem

A population has mean $\mu = 100$ and standard deviation $\sigma = 21$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

**a.** 2

**b.** 3

**c.** 4

**d.** 21

**25** [Foundational] • Target: Sampling distribution of a sample proportion

A population has mean $\mu = 100$ and standard deviation $\sigma = 84$. For samples of size $n = 36$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

**a.** 14

**b.** 13

**c.** 15

**d.** 84

**26** [Foundational] • Target: Comparing sampling variability across sample sizes

A population has mean $\mu = 100$ and standard deviation $\sigma = 140$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

**a.** 140

**b.** 20

**c.** 21

**d.** 19

**27** [Foundational] • Target: Applying the sampling distribution of the sample mean

A population has mean $\mu = 100$ and standard deviation $\sigma = 49$. For samples of size $n = 49$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

**a.** 49

**b.** 8

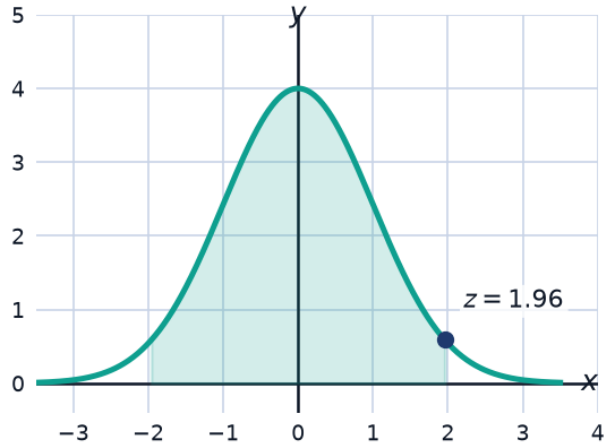
**c.** 7

**d.** 6

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**28** [Foundational] • Target: Sample size and the CLT threshold

A population has a true proportion $p = 0.40$. For samples of size $n = 200$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.




**a.** 0.0346

**b.** 0.0246

**c.** 0.2400

**d.** 0.0446

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Standard error vs. standard deviation

A population has a true proportion $p = 0.80$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

**a.** 0.0566

**b.** 0.1600

**c.** 0.0466

**d.** 0.0666

**30** [Intermediate] • Target: The Central Limit Theorem (CLT)

A population has a true proportion $p = 0.70$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

**a.** 0.0365

**b.** 0.0265

**c.** 0.0165

**d.** 0.2100

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**31** [Intermediate] • Target: Mean and Standard Error of the Sample Mean

A population has a true proportion $p = 0.10$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

**a.** 0.0324

**b.** 0.0524

**c.** 0.0424

**d.** 0.0900

**32** [Intermediate] • Target: Sampling Distribution of Sample Proportion p-hat

A population has a true proportion $p = 0.30$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

**a.** 0.2100

**b.** 0.0365

**c.** 0.0165

**d.** 0.0265

**33** [Intermediate] • Target: Calculations with Sampling Distributions

A population has a true proportion $p = 0.30$. For samples of size $n = 500$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

**a.** 0.0305

**b.** 0.0205

**c.** 0.2100

**d.** 0.0105

**34** [Intermediate] • Target: What is a sampling distribution?

A population has a true proportion $p = 0.20$. For samples of size $n = 50$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

**a.** 0.0566

**b.** 0.1600

**c.** 0.0666

**d.** 0.0466

**35** [Intermediate] • Target: Mean and standard deviation of the sampling distribution

A population has mean $\mu = 112$ and standard deviation $\sigma = 165$ (population shape unknown). For a random sample of $n = 121$, find the probability the sample mean is above $127$, using the Central Limit Theorem.

**a.** 84.13%

**b.** 19.87%

**c.** 11.87%

**d.** 15.87%

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**36** [Intermediate] • Target: The Central Limit Theorem

A population has mean $\mu = 88$ and standard deviation $\sigma = 105$ (population shape unknown). For a random sample of $n = 49$, find the probability the sample mean is below $118$, using the Central Limit Theorem.

**a.** 97.72%

**b.** 2.28%

**c.** 94.72%

**d.** 93.72%

**37** [Intermediate] • Target: Applying the Central Limit Theorem

A pollster samples $n = 1200$ voters, from a population with true proportion $p = 0.70$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.

**a.** 0.7000

**b.** 0.0052

**c.** 0.0132

**d.** 0.0212

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Sampling distribution of a sample proportion • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A pollster samples $n = 800$ voters, from a population with true proportion $p = 0.60$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.

**a.** 0.0253

**b.** 0.6000

**c.** 0.0173

**d.** 0.0093

**39** [Challenging] • Target: Comparing sampling variability across sample sizes • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A population has standard deviation $\sigma = 60$. Find the sample size $n$ needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals $5$.

**a.** 144

**b.** 132

**c.** 12

**d.** 156

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**40** [Challenging] • Target: Applying the sampling distribution of the sample mean • Cross-Lesson: 3.03 The Normal Distribution

A population has standard deviation $\sigma = 152$. Find the sample size $n$ needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals $8$.

**a.** 342

**b.** 19

**c.** 380

**d.** 361

**41** [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

What does the Central Limit Theorem state about the sampling distribution of the sample mean $\bar{x}$?

**a.** When sample size $n$ is sufficiently large ($n \geq 30$), the sampling distribution of $\bar{x}$ is approximately normal, regardless of the population shape

**b.** As $n$ increases, the population becomes normally distributed

**c.** The sample standard deviation approaches the population standard deviation

**d.** Every random sample will have an identical mean

**42** [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Does the Central Limit Theorem apply to the shape of the individual sample data values?

**a.** No, it applies only to the distribution of sample statistics across repeated sampling, not the sample data itself

**b.** Yes, individual observations become normal as $n$ grows

**c.** Only if the data has no outliers

**d.** Yes, this is known as the Law of Large Numbers

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**43** [Challenging] • Target: Mean and Standard Error of the Sample Mean • Cross-Lesson: 1.01 Measures of Center & Spread

When using the formula $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, what condition must be satisfied if sampling without replacement?

**a.** The 10% condition: sample size $n \leq 0.10N$ (population size)

**b.** The Large Counts Condition: $np \geq 10$

**c.** $n \geq 30$

**d.** The population must be symmetric

**44** [Challenging] • Target: Sampling Distribution of Sample Proportion p-hat • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Which conditions justify modeling the sampling distribution of $\hat{p}$ with a normal distribution?

**a.** $np \geq 10$ and $n(1-p) \geq 10$

**b.** $n \geq 30$

**c.** $n \leq 0.10N$

**d.** $p = 0.50$

**45** [Challenging] • Target: Calculations with Sampling Distributions • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A tire manufacturer knows tire lifespan is normal with $\mu = 50000$ miles and $\sigma = 4000$ miles. What is the probability that an SRS of 16 tires has a mean lifespan under 48000 miles?

**a.** $P(Z < -2.0) \approx 0.0228$

**b.** $P(Z < -0.5) \approx 0.3085$

**c.** $P(Z > 2.0) \approx 0.9772$

**d.** 0.0500

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